Transformations of the graph of a function

Four moves — shift, shift, stretch, reflect — and the rule that inside the bracket everything works backwards.

LESSON 22–23 | 2 × 40 MIN | ALGEBRA 10, §1.7 | P1 · 3.1–3.4

LESSON CLOCK

10' | 16' | 16' | 16' | 22' | 10'

Outside the bracket — what you expect	10 min
Inside the bracket — backwards	16 min
Stretches	16 min
Reflections and combinations	16 min
Practice	22 min
Homework	10 min

LEARNING OBJECTIVES

- Sketch y = f(x) + a and y = f(x − a) from the graph of f.
- Sketch y = a·f(x) and y = f(ax).
- Sketch y = −f(x) and y = f(−x).
- Combine two transformations and give them in the correct order.

TEXTBOOK

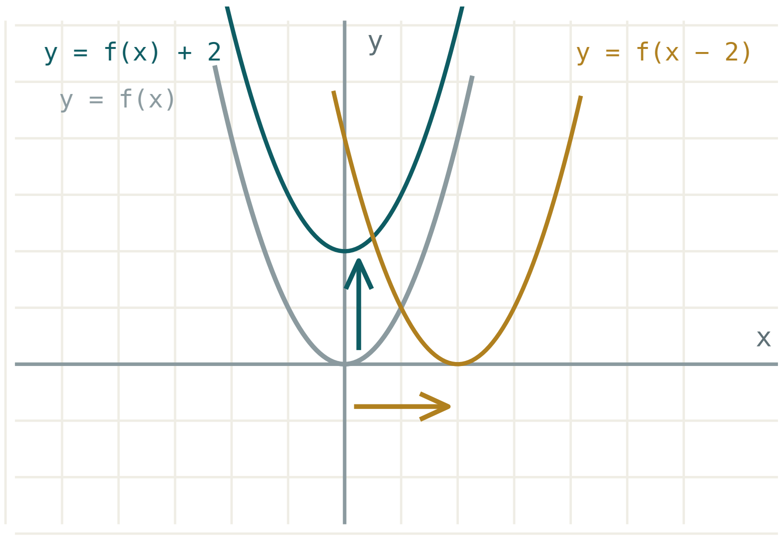
National	pp. 71–80
Cambridge	Pure Mathematics 1, pp. 58–74
Workbook	P1 Exercise 3A–3C

01 Explanation

Outside the bracket behaves

Everything done to $f(x)$ **after** the function has acted moves the graph vertically, in the direction you would guess:

EQUATION	EFFECT	A POINT (P, Q) GOES TO
$y = f(x) + a$	up by a	$(p, q + a)$
$y = a{\cdot}f(x)$	stretch, factor a, from the x-axis	(p, aq)
$y = -f(x)$	reflect in the x-axis	$(p, -q)$



The same curve, shifted. Only the y-coordinates changed.

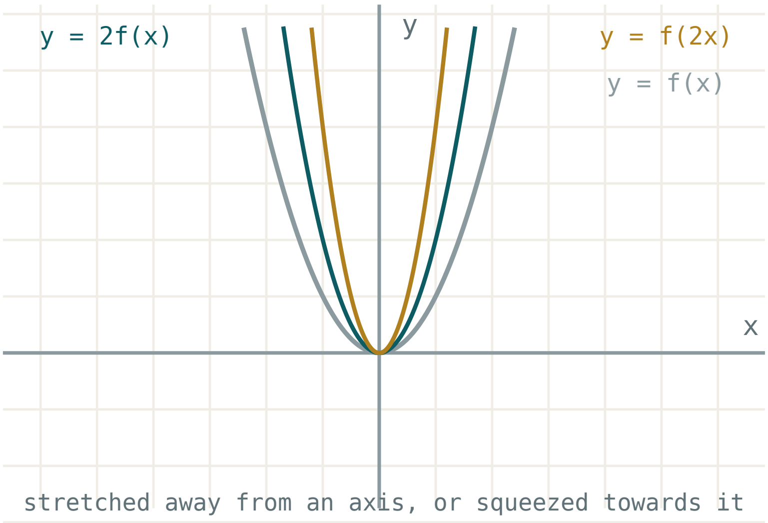
Inside the bracket works backwards

THE ONE RULE TO MEMORISE

Anything done to x **before** the function acts moves the graph horizontally and does the **opposite** of what it looks like. $f(x-3)$ moves the graph **right** 3, not left.

Why: the new graph at $x = 3$ shows the old value at 0. To see the old picture you must travel further along, so the whole graph slides right.

EQUATION	EFFECT	A POINT (P, Q) GOES TO
$y = f(x-a)$	right by a	$(p+a, q)$
$y = f(ax)$	compress, factor $\frac{1}{a}$, towards the y-axis	$(\frac{p}{a}, q)$
$y = f(-x)$	reflect in the y-axis	$(-p, q)$



A stretch of factor 2 from the x-axis, and a compression of factor ½ towards the y-axis. Different curves.

> *F(2X)* IS NOT A STRETCH BY 2
>
> It is a compression: every x-coordinate is **halved**. $f(\frac{x}{2})$ is the one that stretches by 2.

Combining, and the order

$y = 3f(x - 2) + 1$ is three transformations. Outside-the-bracket operations are applied in ordinary BIDMAS order — stretch first, then shift:

1. $f(x - 2)$: right 2.
2. $3f(x - 2)$: stretch factor 3 from the x-axis.
3. $3f(x - 2) + 1$: up 1.

> ORDER CHANGES THE ANSWER
>
> Stretch-then-shift and shift-then-stretch give different curves. $3f(x) + 1$ and $3[f(x) + 1]$ are not the same: the second is $3f(x) + 3$.

Completed-square form is exactly this idea for quadratics: $y = (x - 3)^2 + 2$ says “take $y = x^2$, move it right 3 and up 2” — which is why the vertex is $(3, 2)$.

02 Worked examples

EXAMPLE 1 **The point $(2, 5)$ is on $y = f(x)$. Where is it on $y = 2f(x + 1) - 3$?**

1. $x + 1$ inside: move left 1.
   x-coordinate $2 - 1 = 1$.
2. Stretch by 2: $5 \cdot 2 = 10$.
3. Down 3: $10 - 3 = 7$.

ANSWER

$(1, 7)$

EXAMPLE 2 **Describe the transformation taking $y = x^2$ to $y = (x + 4)^2 - 9$.**

1. $(x + 4)$ inside: left 4.
   Backwards, as always.
2. $-\ 9$ outside: down 9.
3. Vertex moves from $(0,0)$ to $(-4, -9)$.

ANSWER

Translation left 4 and down 9

EXAMPLE 3 **Write $y = x^2 - 10x + 18$ as a transformation of $y = x^2$.**

1. Complete the square.
   $(x - 5)^2 - 25 + 18$

2. $= (x - 5)^2 - 7$

3. Read off the two moves.

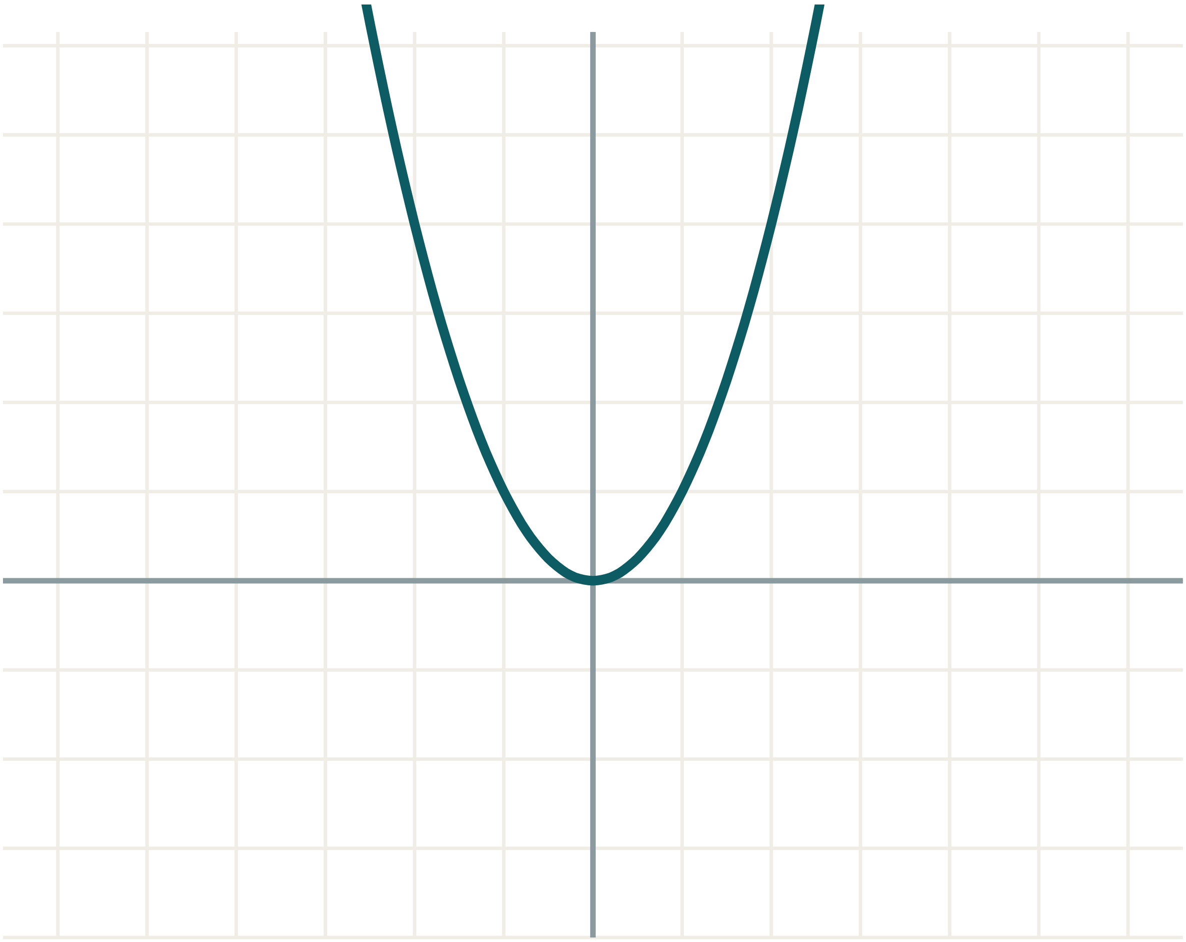
ANSWER

Right 5, down 7 — vertex $(5, -7)$

03 Interactive model

Drag a, b and c and watch which coordinates move.

y = a·f(x − b) + c



a **1** | b **0** | c **0** | equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Translation	Parallel ko‘chirish	Параллельный перенос
Vertical shift	Vertikal siljish	Вертикальный сдвиг
Horizontal shift	Gorizontal siljish	Горизонтальный сдвиг
Stretch	Cho‘zish	Растяжение
Compression	Siqish	Сжатие
Scale factor	Cho‘zish koeffitsienti	Коэффициент растяжения
Reflection	Simmetriya	Отражение
Invariant point	O‘zgarmas nuqta	Неподвижная точка
Image of a point	Nuqta tasviri	Образ точки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$y = f(x) + 5$ moves the graph:

- up 5
- down 5
- left 5
- right 5

$y = f(x - 5)$ moves the graph:

- left 5
- right 5
- up 5
- down 5

$y = f(3x)$ is a:

- stretch factor 3
- compression factor $\frac{1}{3}$
- shift right 3
- reflection

$y = -f(x)$ reflects in:

- *the y-axis*
- *the x-axis*
- $y = x$
- the origin

The vertex of $y = (x + 1)^2 + 6$ is:

- $(1, 6)$
- $(-1, 6)$
- $(-1, -6)$
- $(1, -6)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Describe $y = f(x) + 3$

   ANSWER up 3

2. Describe $y = f(x) - 7$

   ANSWER down 7

3. Describe $y = f(x - 2)$

   ANSWER right 2

4. Describe $y = f(x + 6)$

   ANSWER left 6

5. Describe $y = 4f(x)$

   ANSWER stretch factor 4 from the x-axis

6. Describe $y = -f(x)$

   ANSWER reflect in the x-axis

7. Vertex of $y = (x - 1)^2$

   ANSWER $(1, 0)$

Medium · the standard question

7 PROBLEMS

1. Vertex of $y = (x + 3)^2 - 5$

   ANSWER $(-3, -5)$

2. $(4, 2)$ on $y = f(x)$; find it on $y = f(x) + 6$

   ANSWER $(4, 8)$

3. Same point on $y = f(x - 1)$

   ANSWER $(5, 2)$

4. Same point on $y = 3f(x)$

   ANSWER $(4, 6)$

5. Same point on $y = f(2x)$

   ANSWER $(2, 2)$

6. Write $y = x^2 + 6x + 4$ as a transformation of $y = x^2$

   ANSWER left 3, down 5

7. Describe $y = f(-x)$

   ANSWER reflect in the y-axis

Hard · stretch and reason

7 PROBLEMS

1. $(3, -1)$ on $y = f(x)$; find it on $y = 2f(x + 3) - 4$

   ANSWER $(0, -6)$

2. Describe $y = -2f(x) + 1$ in order

   ANSWER stretch 2, reflect in x-axis, up 1

3. Write $y = 2x^2 - 12x + 20$ in the form $a(x - h)^2 + k$

   ANSWER $2(x - 3)^2 + 2$

4. What single transformation maps $y = x^2$ to $y = 4x^2$?

   ANSWER stretch factor 4 from the x-axis (or compression $\frac{1}{2}$ towards the y-axis)

5. Find a, b with $f(x) = (x - a)^2 + b$ passing through $(0, 10)$ and vertex on $x = 3$

   ANSWER $a = 3, b = 1$

6. Which points of $y = f(x)$ are unmoved by $y = -f(x)$?

   ANSWER those with $y = 0$

7. Which points are unmoved by $y = f(-x)$?

   ANSWER those with $x = 0$

07 Homework

Homework — 5 tasks

Sketch, do not only describe. One axis set per question is enough.

1. Sketch $y = x^2$, $y = (x - 2)^2$ and $y = (x - 2)^2 + 3$ on one set of axes.
2. Describe the transformation from $y = x^2$ to $y = x^2 - 4x + 1$.
3. $(-2, 7)$ is on $y = f(x)$. Find its image on $y = 3f(x + 2) - 1$.
4. Explain in your own words why $f(x - 3)$ moves the graph right.

5. Sketch $y = -f(x)$ and $y = f(-x)$ for a curve of your choice and say how they differ.